

Lab8_Q1: Library Loan & Late Fee System

You are required to design a **minimal but fully functional library loan management system** using **exactly two classes**:

1. **Book**
2. **Library**

System Features to Implement

1. Book Borrowing

- If a book is **available**, the requesting user should receive it immediately.
- If the book is **currently borrowed**, the user must be **added to the waitlist** for that specific title.

2. Multi-Book Borrowing (Simultaneous Loans)

- A user may borrow **more than one book at the same time**.
- The system must correctly track multiple active loans per user.

3. Book Returning & Waitlist Promotion

- When a user returns a book:
 - If the **waitlist is empty**, the book simply becomes available.
 - If the **waitlist is not empty**, the next user in line must automatically receive the book.

4. Late Fee Calculation

You must implement a billing logic based on how late the book is returned:

- Maximum allowed loan period: **14 days**
- Daily late charge: **50 TL**
- If returned on or before day 14 → no fee.

5. Library-Level Tracking

Your `Library` class must maintain:

- All book objects
- A record of which books are currently borrowed by which users
- Correct updates when books are returned and auto-transferred to waitlisted users

Input	Output
lib.borrow_book("Alice", "1984")	Alice borrowed 1984 waitlist=[]
lib.borrow_book("Bob", "1984")	1984 unavailable, Bob added to waitlist waitlist=['Bob']
lib.borrow_book("Charlie", "1984")	1984 unavailable, Charlie added to waitlist waitlist=['Bob', 'Charlie']
lib.borrow_book("Dave", "1984")	1984 unavailable, Dave added to waitlist waitlist=['Bob', 'Charlie', 'Dave']
lib.borrow_book("Alice", "Dune")	Alice borrowed Dune waitlist=[]
lib.borrow_book("Eve", "Dune")	Dune unavailable, Eve added to waitlist waitlist=['Eve']
print(lib.list_user_books("Alice"))	['1984', 'Dune']
print(lib.list_user_books("Bob"))	[]
print(lib.list_user_books("Charlie"))	[]
print(lib.list_user_books("Dave"))	[]
print(lib.list_user_books("Eve"))	[]
lib.return_book("Alice", "1984", 35)	Alice returned 1984, fee=42, next=Bob waitlist=['Charlie', 'Dave']
lib.return_book("Bob", "1984", 22)	Bob returned 1984, fee=16, next=Charlie waitlist=['Dave']
lib.return_book("Charlie", "1984", 10)	Charlie returned 1984, fee=0, next=None waitlist=[]
print(lib.list_user_books("Dave"))	['Dave']